

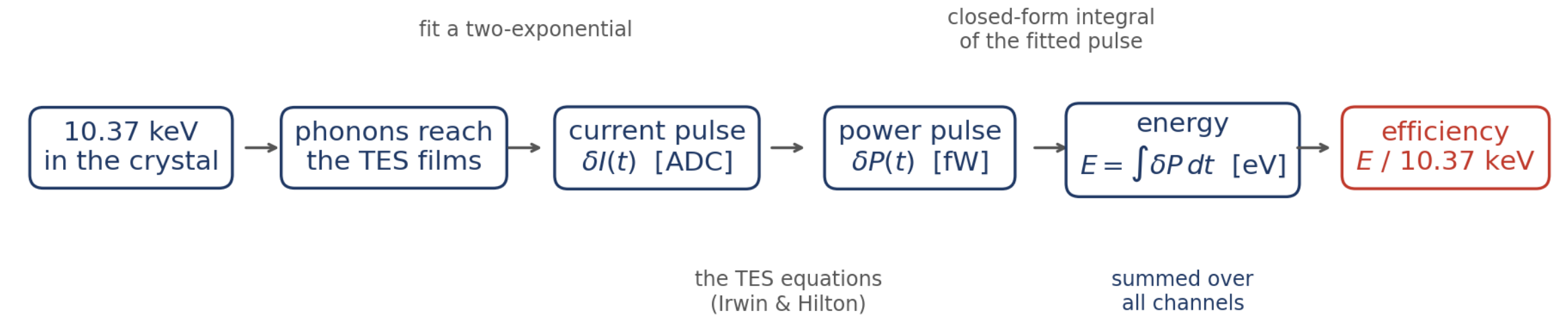
# Phonon collection efficiency of Z7

# Z7 的声子收集效率

Zhiheng Li · University of Minnesota · SuperCDMS SNOLAB Run 4 · September 2026  
github.com/ZhihengLi0/urop → snolab/collection\_efficiency/

李知恒 · 明尼苏达大学 · SuperCDMS SNOLAB Run 4 · 2026 年 9 月  
github.com/ZhihengLi0/urop → snolab/collection\_efficiency/

# The Main Idea

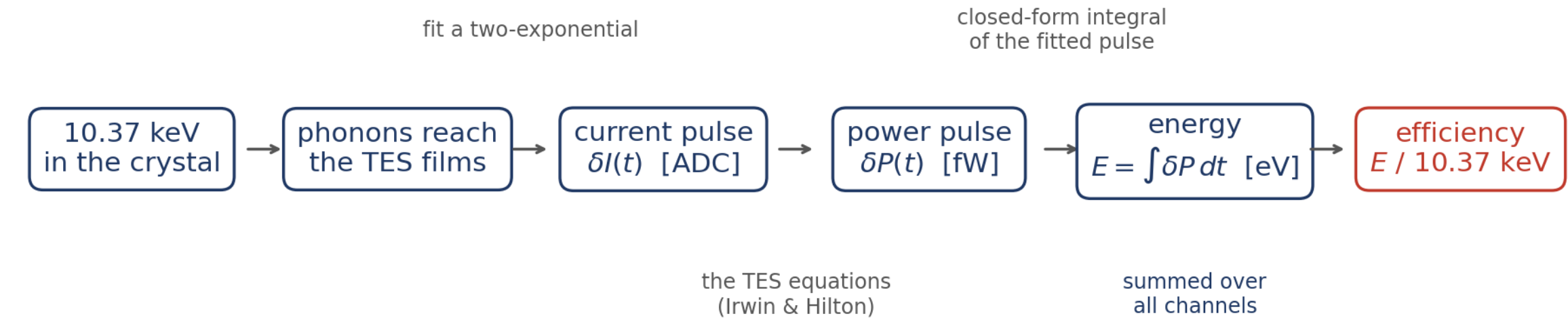


## For detectors operated at 0 V (no NTL effect):

Collection efficiency = energy absorbed by the TESs, summed over all channels ÷ the 10.37 keV of the event

Sample: Z7, the K-line events of the Ge activation run, from SNOLAB R4 (2207 events, 30 series, all 11 channels).

# 主要思路



## 对于在 0 V 下工作的探测器（没有 NTL 效应）：

收集效率 = TES 吸收的能量（所有通道求和） ÷ 事件放进晶体的 10.37 keV

样本：Z7，Ge 活化的 K 线事件，来自 SNOLAB R4（2207 个事件，30 个 series，11 个通道）。

# The formula, and the two numbers it needs

Method 1 of the CDMS collection-efficiency note

$$\delta P(t) = I_0 (R_L - R_0) \delta I(t) + 2 R_L (\delta I(t))^2$$

linear term

quadratic term

$$E = \int \delta P(t) dt = I_0 (R_L - R_0) A (\tau_f - \tau_r) + 2 R_L A^2 \left( \frac{\tau_f + \tau_r}{2} - \frac{2 \tau_f \tau_r}{\tau_f + \tau_r} \right)$$

for the fitted pulse  $\delta I(t) = A [ e^{-t/\tau_f} - e^{-t/\tau_r} ]$  integrated from  $-\infty$  to  $+\infty$ : zero before it starts, decays to zero after

Z7 PBS1:  $I_0 = -18.07 \mu A, \quad R_0 = 44.16 \text{ m}\Omega, \quad R_L = R_p + R_{sh} = 14.24 \text{ m}\Omega + 5.00 \text{ m}\Omega = 19.24 \text{ m}\Omega$

linear coefficient:  $I_0 (R_L - R_0) = (-18.07 \mu A) \times (19.24 \text{ m}\Omega - 44.16 \text{ m}\Omega) = 0.45 \mu V$

quadratic coefficient:  $2 R_L = 2 \times 19.24 \text{ m}\Omega = 38.5 \text{ m}\Omega = 0.0385 \Omega$

$\delta P$  = the change of Joule heating when the current moves by  $\delta I$ : a linear term and a small quadratic term.

$I_0$ : TES current at the bias point ·  $R_0$ : TES resistance at the bias point ·  $R_L = R_p + R_{sh}$ : load resistance ( $R_p$  parasitic,  $R_{sh}$  shunt) ·  $\delta I$ : change in current, from the trace

The note also gives Method 2:  $\delta P = I_0 (R_L - R_0) \cdot \delta I - R_L \cdot (\delta I)^2$  in our sign convention — same linear term, only the quadratic term differs, 1.7 % less energy (PBS1, median of 15 events).

Source: Antoine Rehberg, CDMS wiki, Collection Efficiency Analysis — [confluence.slac.stanford.edu/spaces/CDMS/pages/542577254](https://confluence.slac.stanford.edu/spaces/CDMS/pages/542577254)

# 公式和它需要的两个数

采用 CDMS 收集效率文档里的 Method 1

$$\delta P(t) = I_0 (R_L - R_0) \delta I(t) + 2 R_L (\delta I(t))^2$$

linear term

quadratic term

$$E = \int \delta P(t) dt = I_0 (R_L - R_0) A (\tau_f - \tau_r) + 2 R_L A^2 \left( \frac{\tau_f + \tau_r}{2} - \frac{2 \tau_f \tau_r}{\tau_f + \tau_r} \right)$$

for the fitted pulse  $\delta I(t) = A [ e^{-t/\tau_f} - e^{-t/\tau_r} ]$  integrated from  $-\infty$  to  $+\infty$ : zero before it starts, decays to zero after

Z7 PBS1:  $I_0 = -18.07 \mu A, \quad R_0 = 44.16 \text{ m}\Omega, \quad R_L = R_p + R_{sh} = 14.24 \text{ m}\Omega + 5.00 \text{ m}\Omega = 19.24 \text{ m}\Omega$

linear coefficient:  $I_0 (R_L - R_0) = (-18.07 \mu A) \times (19.24 \text{ m}\Omega - 44.16 \text{ m}\Omega) = 0.45 \mu V$

quadratic coefficient:  $2 R_L = 2 \times 19.24 \text{ m}\Omega = 38.5 \text{ m}\Omega = 0.0385 \Omega$

$\delta P$  = 电流变化  $\delta I$  之后焦耳热的变化量：一个线性项，加一个很小的二次项。

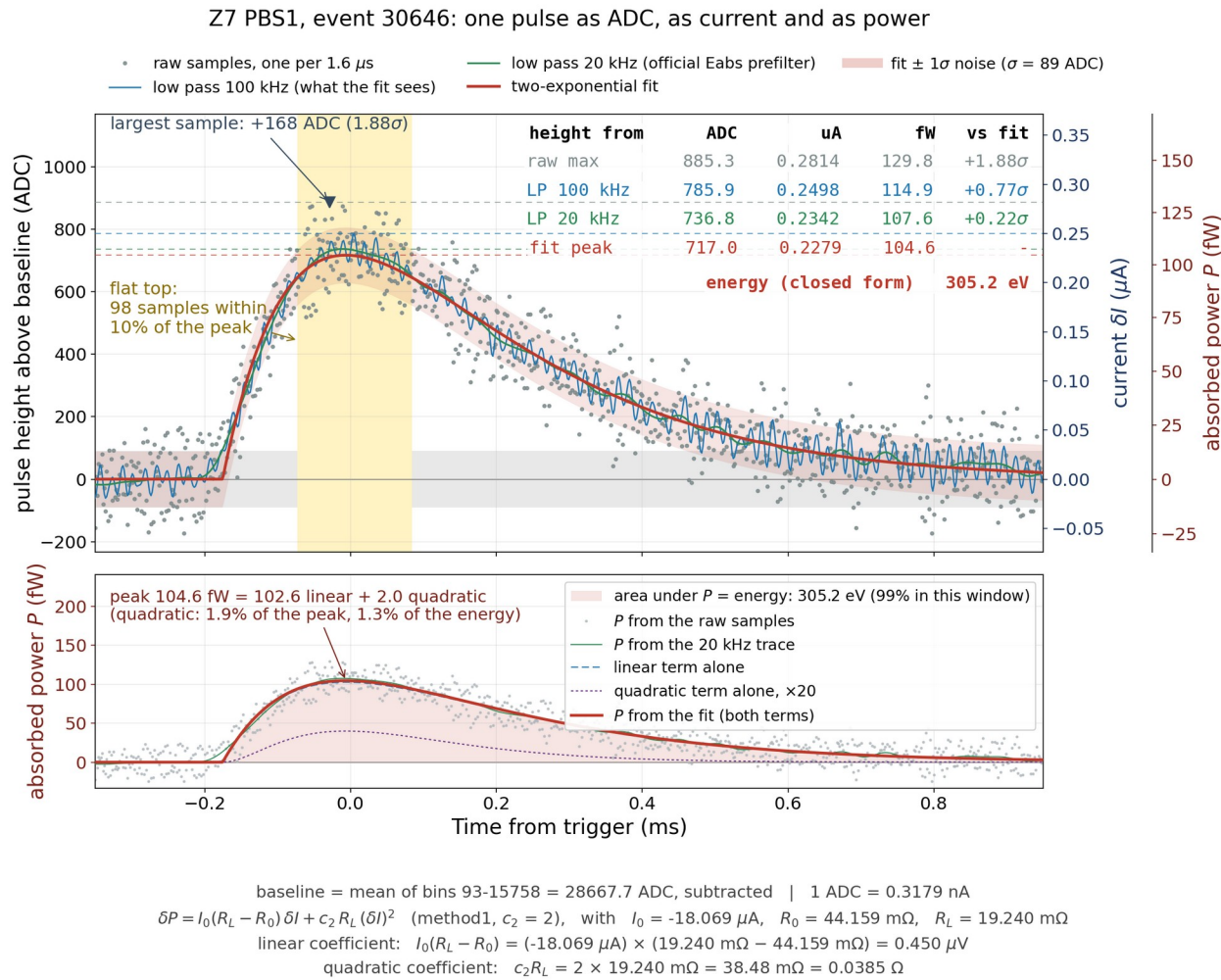
$I_0$ ：工作点上流过 TES 的电流 ·  $R_0$ ：工作点上 TES 的电阻 ·  $R_L = R_p + R_{sh}$ ：负载电阻（ $R_p$  寄生电阻， $R_{sh}$  分流电阻）·  $\delta I$ ：电流的变化，来自波形

文档里还有一个 Method 2：换成我们的符号是  $\delta P = I_0 (R_L - R_0) \cdot \delta I - R_L \cdot (\delta I)^2$  —— 线性项相同，只有二次项不同，能量低 1.7%（PBS1，15 个事件的中位数）。

来源：Antoine Rehberg 的 CDMS wiki 页面 Collection Efficiency Analysis — [confluence.slac.stanford.edu/spaces/CDMS/pages/542577254](https://confluence.slac.stanford.edu/spaces/CDMS/pages/542577254)

# Example: one pulse

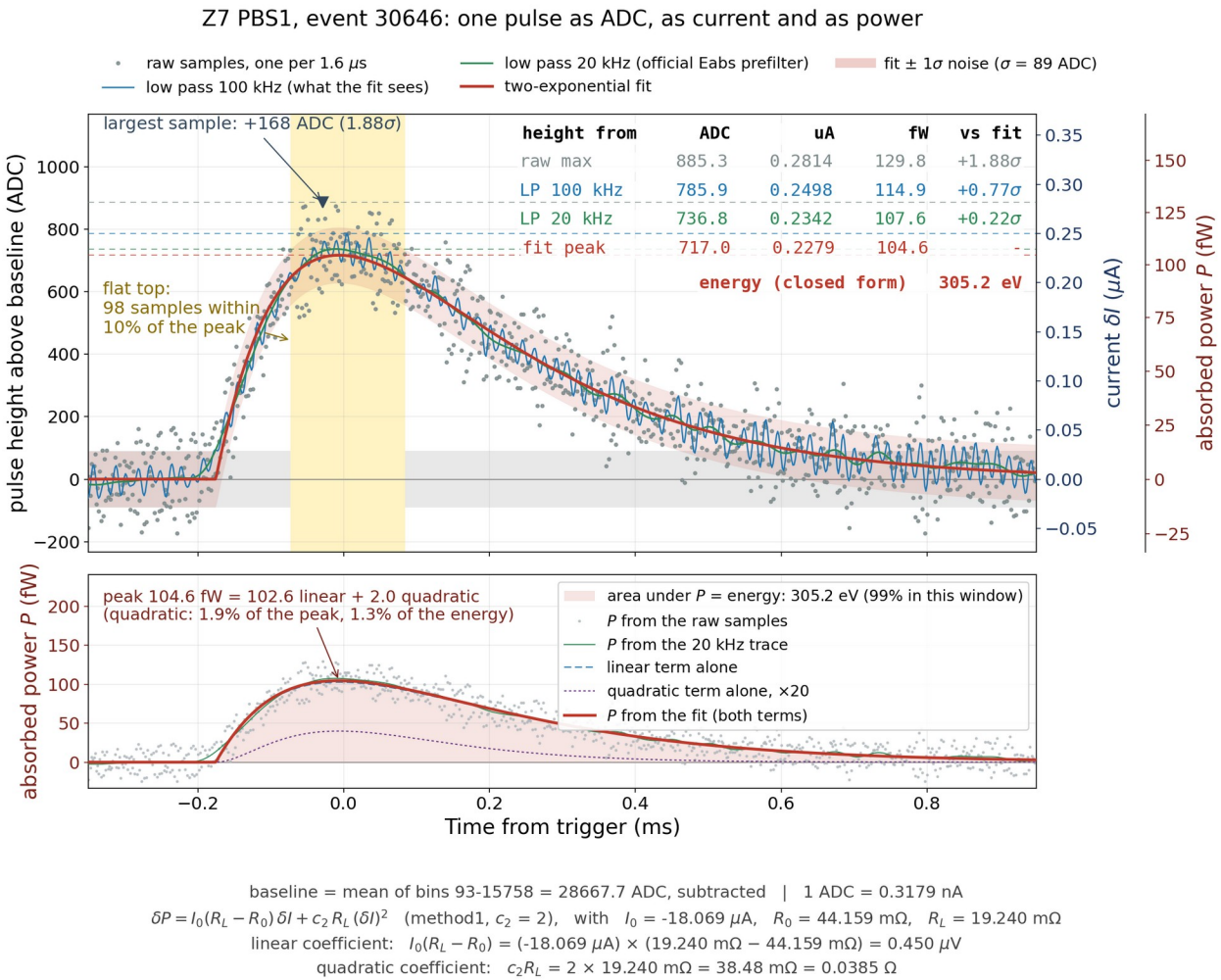
Z7 PBS1, event 30646: the same trace as ADC counts, as current, as power



The two-exponential is fitted to the 100 kHz low-passed trace (blue), not to the raw samples; the raw samples are only drawn.

# 例子：一个脉冲

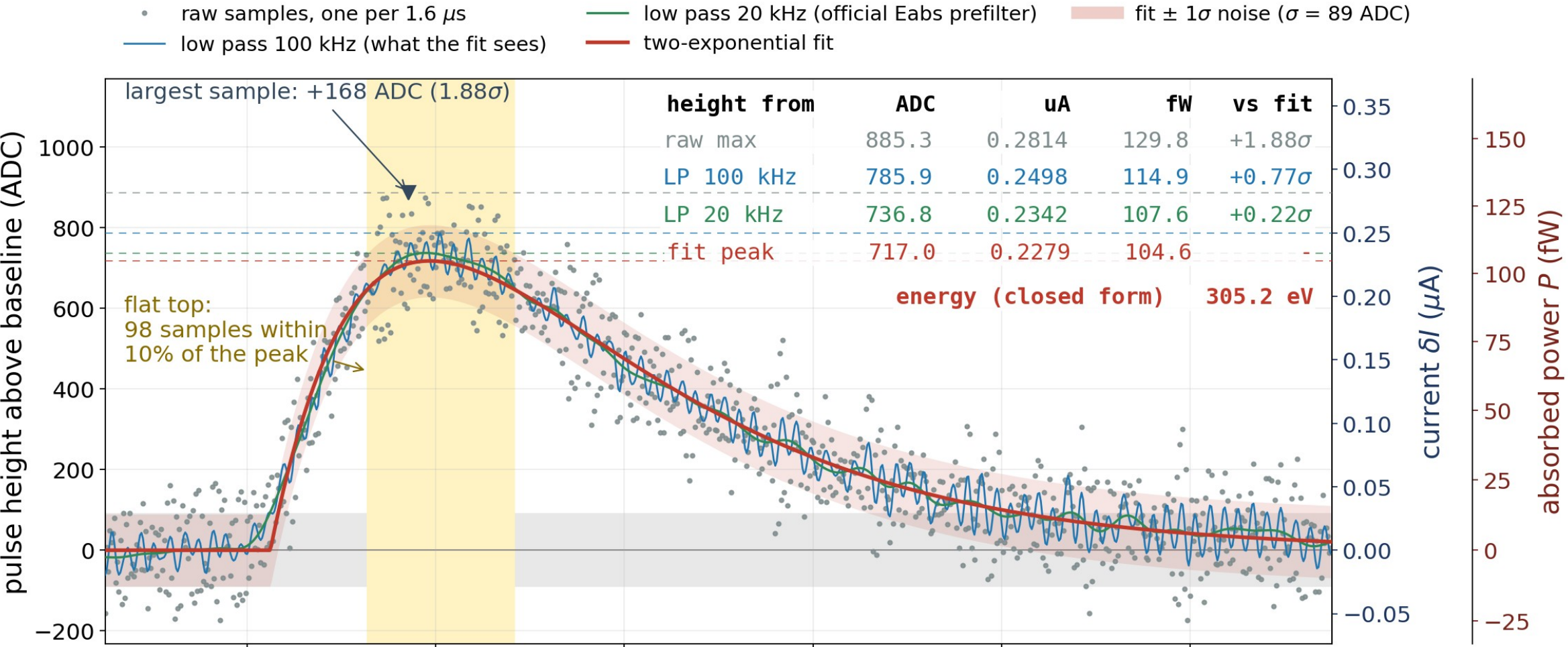
Z7 PBS1 , 事件 30646 : 同一条波形，读成 ADC 计数、读成电流、读成功率



双指数拟合用的是 100 kHz 低通之后的波形（蓝线），不是原始采样点；原始采样点只是画出来对照。

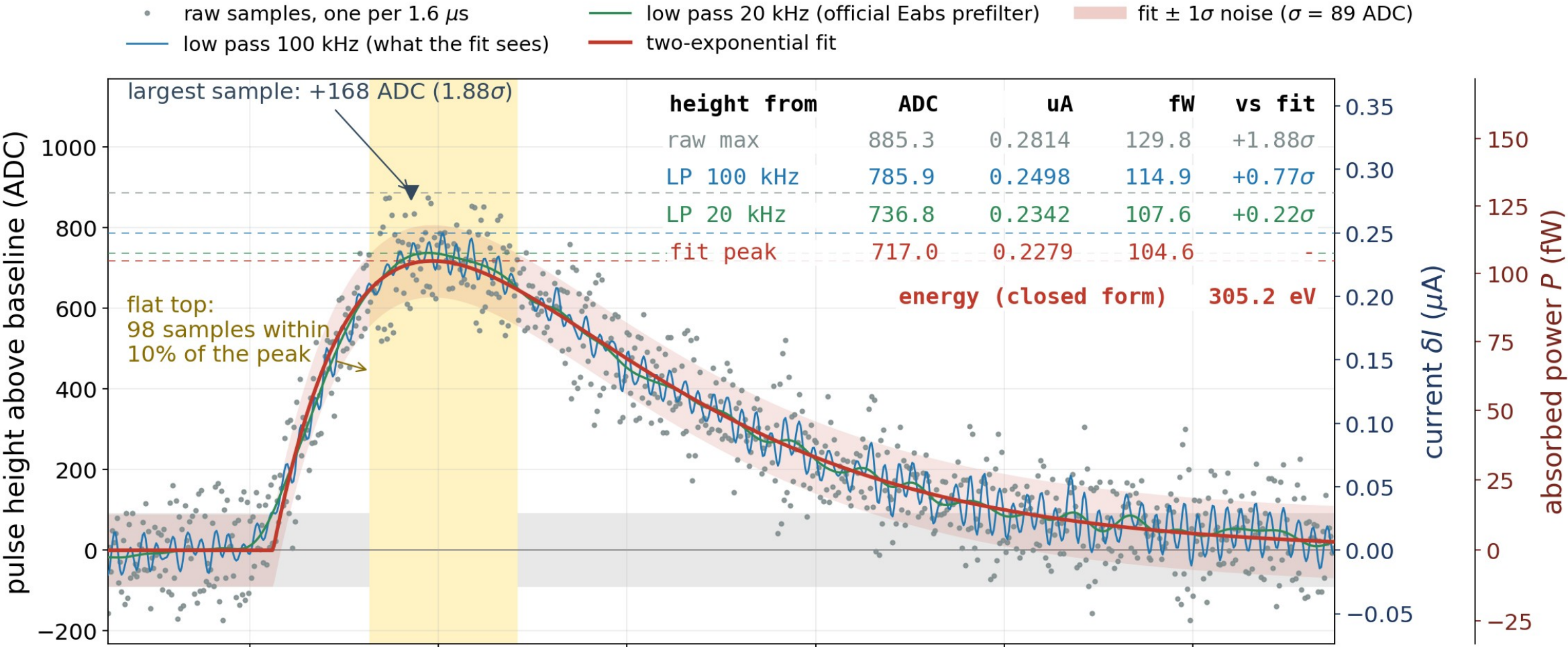
# The height: take it from the fit, not from the largest sample

Z7 PBS1, event 30646: one pulse as ADC, as current and as power

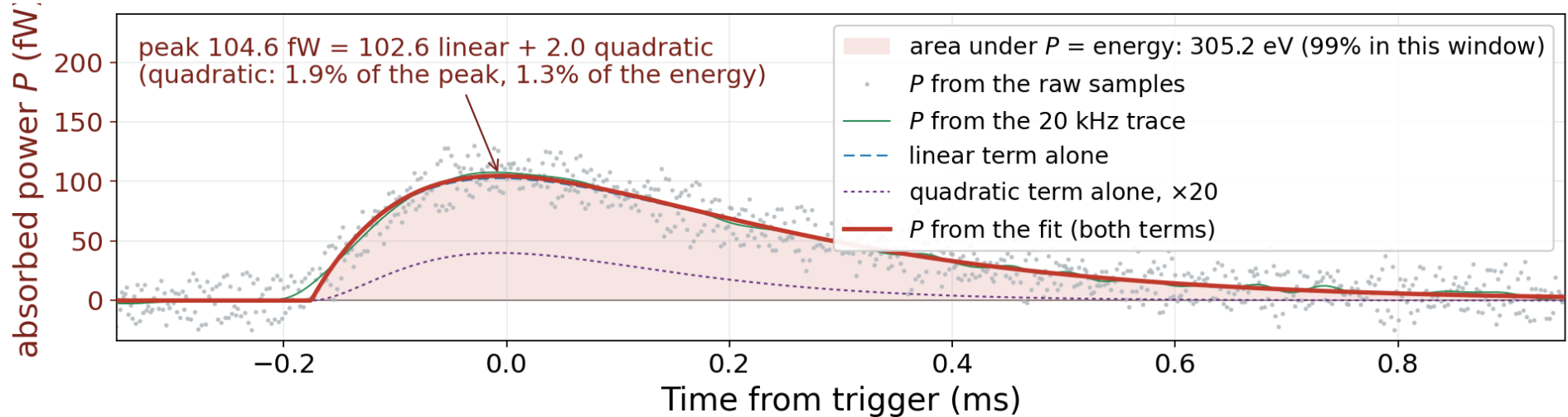


# 脉冲高度：从拟合取，不从最大的采样点取

Z7 PBS1, event 30646: one pulse as ADC, as current and as power



# The power pulse, and its area is the energy



baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA

$\delta P = I_0(R_L - R_0) \delta I + c_2 R_L (\delta I)^2$  (method1,  $c_2 = 2$ ), with  $I_0 = -18.069 \mu\text{A}$ ,  $R_0 = 44.159 \text{ m}\Omega$ ,  $R_L = 19.240 \text{ m}\Omega$

linear coefficient:  $I_0(R_L - R_0) = (-18.069 \mu\text{A}) \times (19.240 \text{ m}\Omega - 44.159 \text{ m}\Omega) = 0.450 \mu\text{V}$

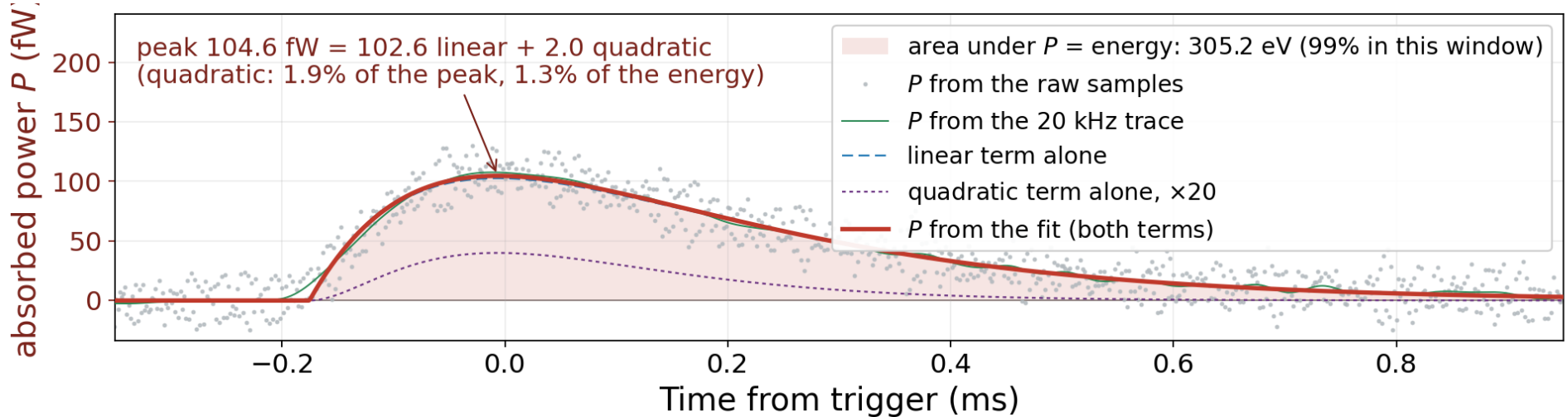
quadratic coefficient:  $c_2 R_L = 2 \times 19.240 \text{ m}\Omega = 38.48 \text{ m}\Omega = 0.0385 \Omega$

Red: the fitted current put through the formula. Blue dashed: the linear term alone. Purple: the quadratic term,  $\times 20$ .

**The quadratic term is 1.9 % of the peak — the power pulse has the shape of the current pulse.**

**The shaded area is the energy this channel absorbed: 305 eV.**

# 功率脉冲，它的面积就是能量



baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA

$\delta P = I_0(R_L - R_0) \delta I + c_2 R_L (\delta I)^2$  (method1,  $c_2 = 2$ ), with  $I_0 = -18.069 \mu\text{A}$ ,  $R_0 = 44.159 \text{ m}\Omega$ ,  $R_L = 19.240 \text{ m}\Omega$

linear coefficient:  $I_0(R_L - R_0) = (-18.069 \mu\text{A}) \times (19.240 \text{ m}\Omega - 44.159 \text{ m}\Omega) = 0.450 \mu\text{V}$

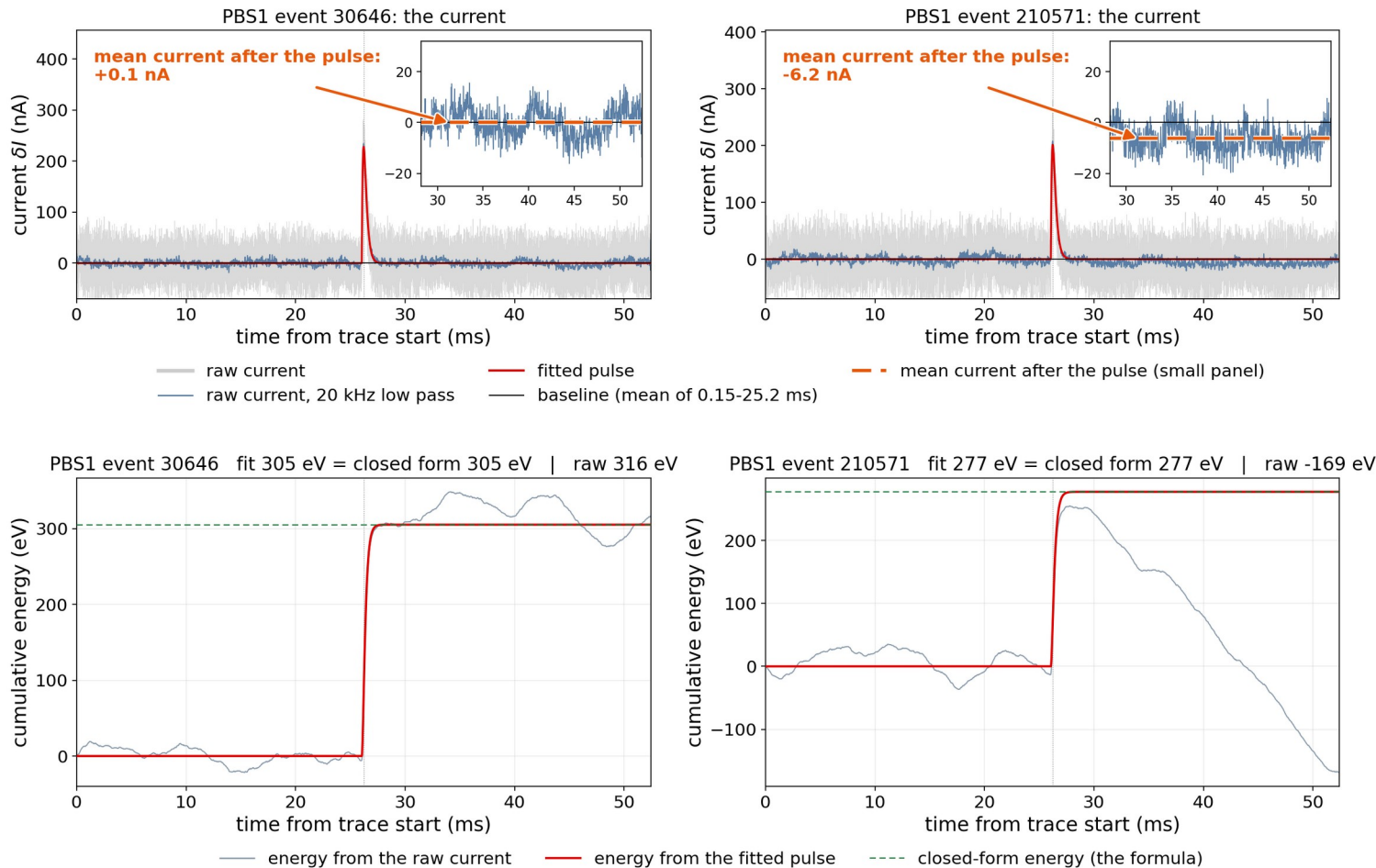
quadratic coefficient:  $c_2 R_L = 2 \times 19.240 \text{ m}\Omega = 38.48 \text{ m}\Omega = 0.0385 \Omega$

红线：拟合出的电流代进公式。蓝色虚线：只有线性项。紫色点线：只有二次项，放大 20 倍。

**二次项只占峰值的 1.9% —— 功率脉冲的形状和电流脉冲一样。**

**粉色面积就是这个通道吸收的能量：305 eV。**

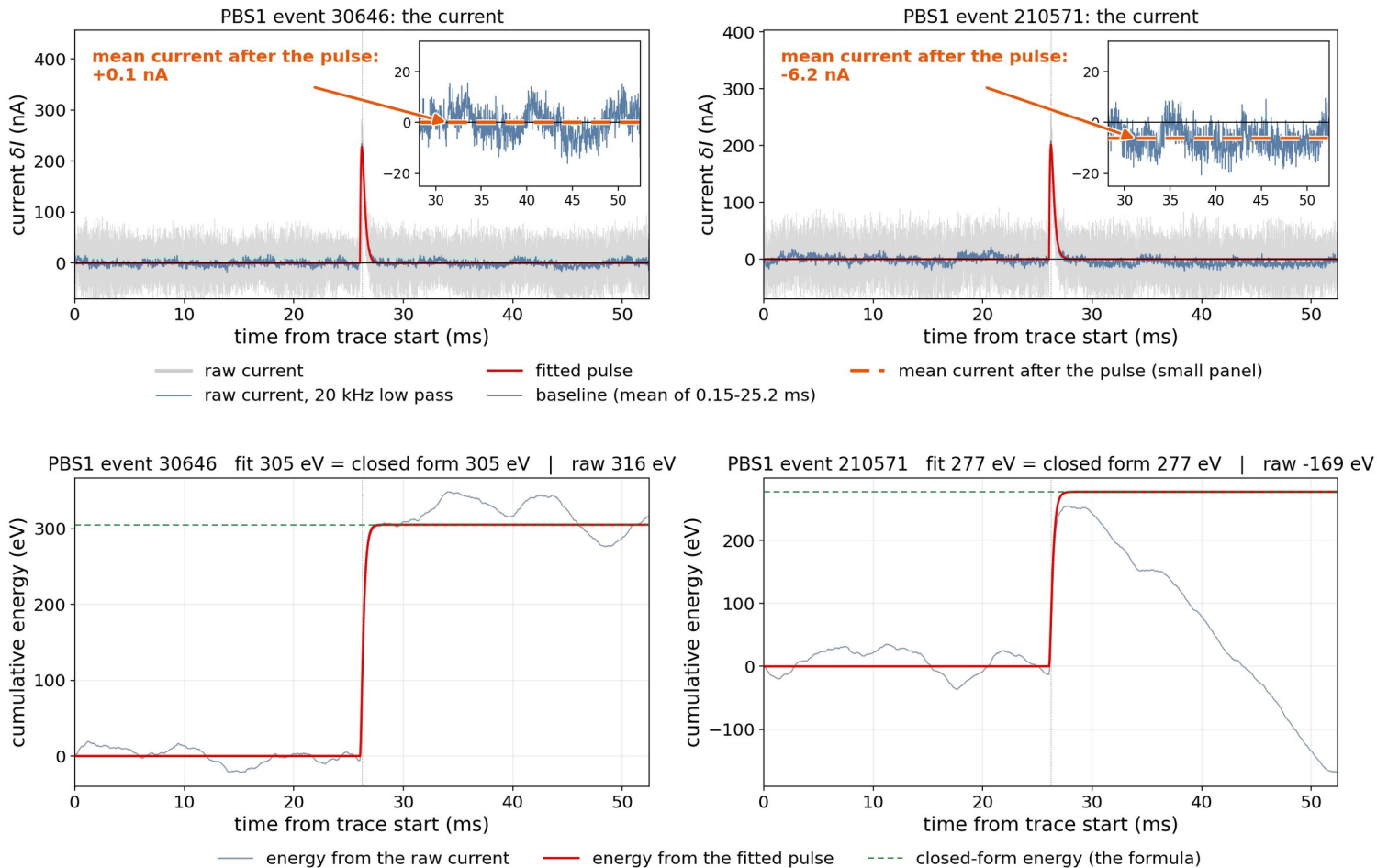
# Why we integrate the fit, not the raw data



Top: the raw current of the two events; the small panel zooms in on the baseline after the pulse. Bottom: the energy accumulated from the raw current (grey) and from the fit (red).

Right: after the pulse the current sits 6 nA below its baseline — hidden in 28 nA of noise, yet over 24 ms it is -420 eV: raw -169 eV, fit 277 eV.

# 为什么对拟合积分，不是对原始数据

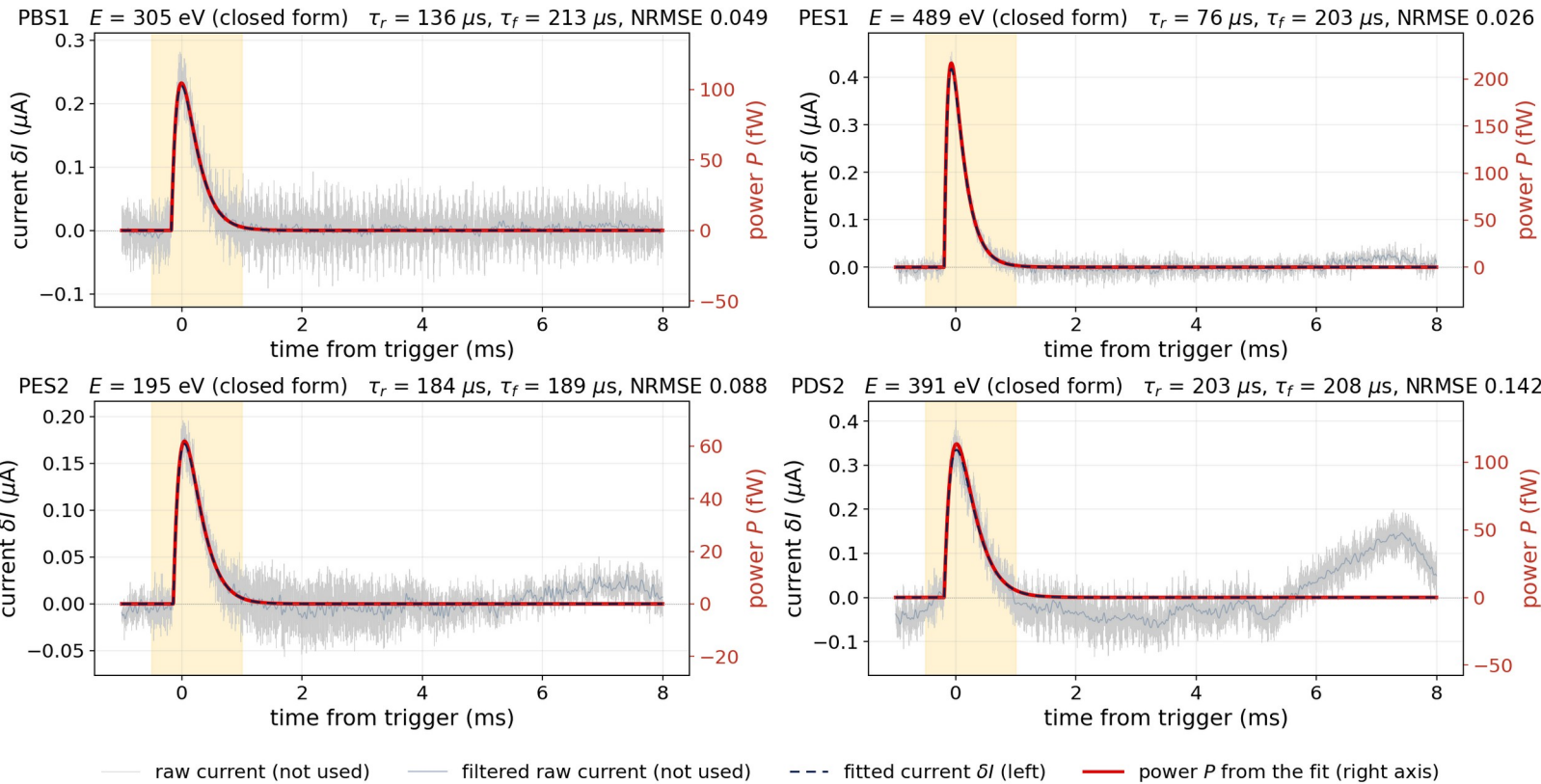


上：这两个事件的原始电流；右上角小图把脉冲之后的基线放大。下：由原始电流累计的能量（灰）和由拟合累计的能量（红）。

右：脉冲过后电流比基线低 6 nA —— 埋在 28 nA 的噪声里看不出来，但 24 ms 积下来就是 -420 eV：原始 -169 eV，拟合 277 eV。

# The same event in its other channels: they add up to 33 %

Z7 event 30646; four of the eleven channels



## energy per channel, event 30646

### side 1 2062 eV (60 %)

|      |        |
|------|--------|
| PAS1 | 303 eV |
| PBS1 | 305 eV |
| PCS1 | 227 eV |
| PDS1 | 378 eV |
| PES1 | 489 eV |
| PFS1 | 359 eV |

### side 2 1357 eV (40 %)

|      |        |
|------|--------|
| PAS2 | 267 eV |
| PBS2 | 261 eV |
| PCS2 | 243 eV |
| PDS2 | 391 eV |
| PES2 | 195 eV |

total 3419 eV = 33.0 % of 10.37 keV

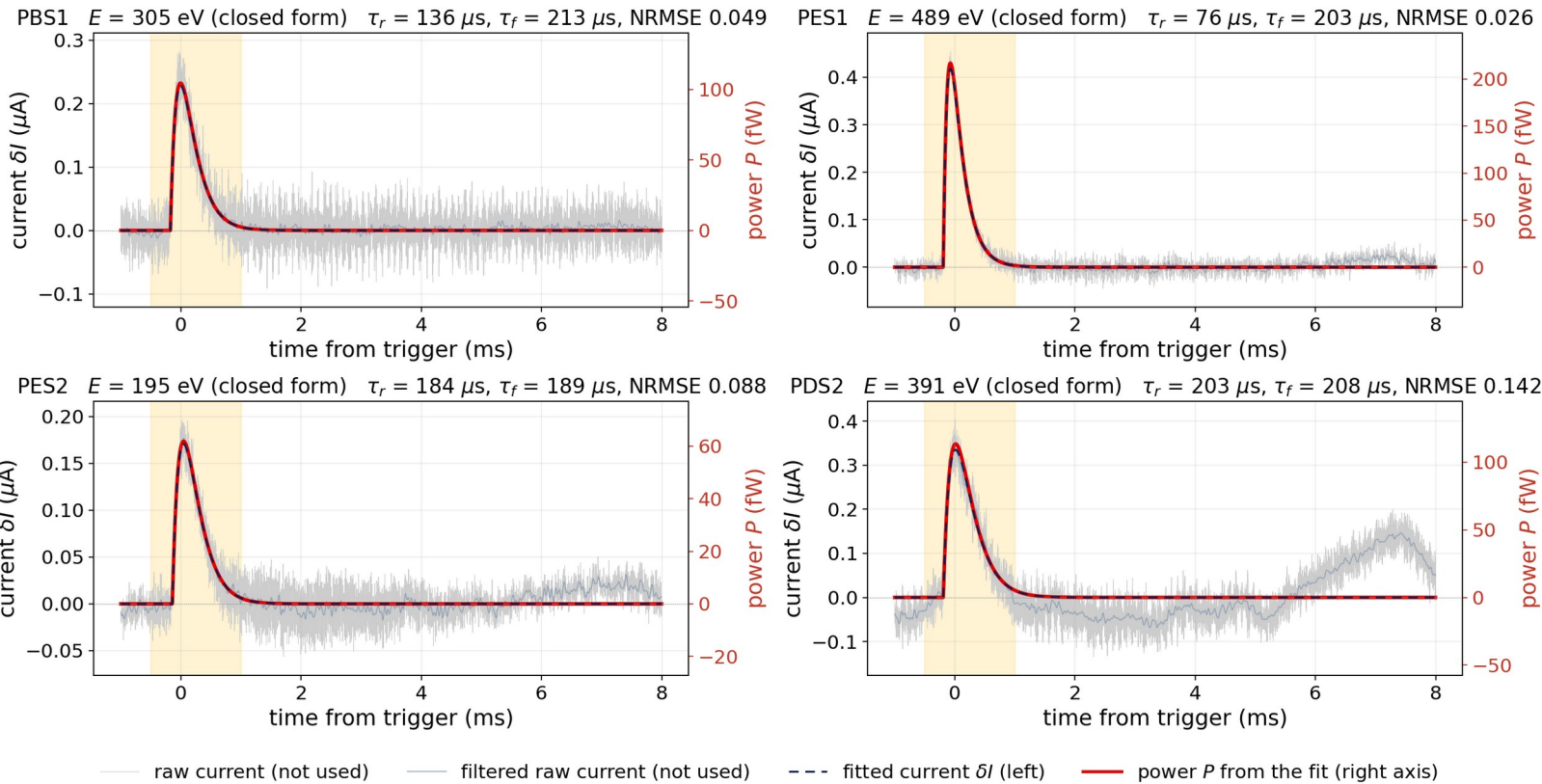
PFS2 is not read out on Z7, so side 2 has 5 channels  
the channels also differ from each other; see the backup

The channels do not share the energy equally: 60 % of it lands on side 1, and PES1 alone has 489 eV against 195 eV in PES2.

Summed over the eleven channels: 3419 eV = 33.0 % of 10.37 keV. PDS2 (bottom right) carries a slow swing.

# 同一个事件的其他通道：加起来是 33%

Z7 事件 30646 ; 11 个通道里的 4 个



## 事件 30646 各通道的能量

### 第 1 面 2062 eV (60 %)

|      |        |
|------|--------|
| PAS1 | 303 eV |
| PBS1 | 305 eV |
| PCS1 | 227 eV |
| PDS1 | 378 eV |
| PES1 | 489 eV |
| PFS1 | 359 eV |

### 第 2 面 1357 eV (40 %)

|      |        |
|------|--------|
| PAS2 | 267 eV |
| PBS2 | 261 eV |
| PCS2 | 243 eV |
| PDS2 | 391 eV |
| PES2 | 195 eV |

合计 3419 eV = 33.0 % of 10.37 keV

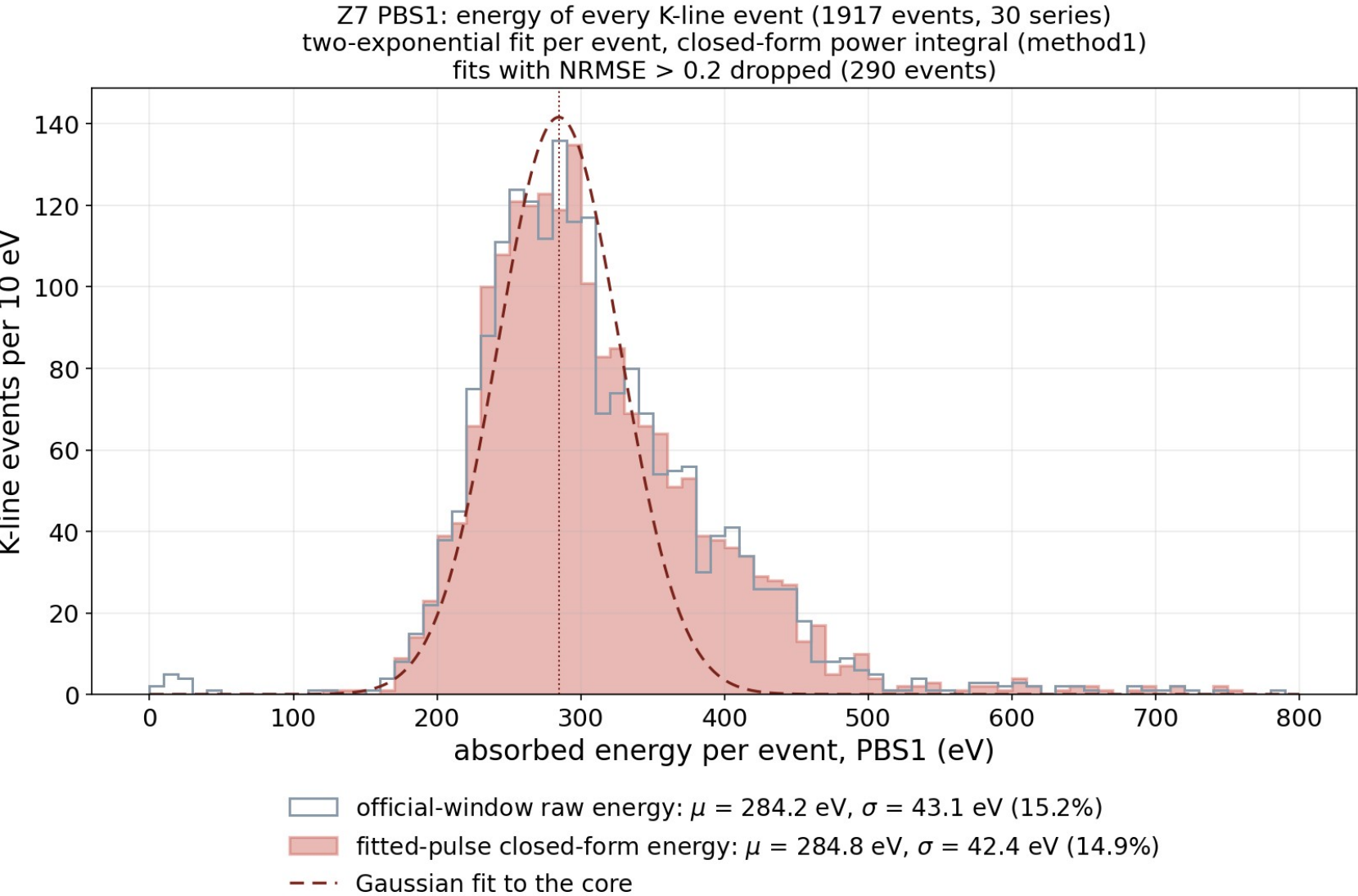
Z7 的 PFS2 没有读出，所以第 2 面只有 5 个通道  
通道之间本身也有固定差别，见 backup 那一页

各通道分到的能量很不均匀：60% 落在第 1 面；PES1 一个通道就有 489 eV，而 PES2 只有 195 eV。

11 个通道加起来：3419 eV = 10.37 keV 的 33.0%。 PDS2（右下）有一个缓慢的大起伏。

# Now every event: one channel gives a wide, lopsided distribution

PBS1, 1917 K-line events, one fit and one integral each



Red: from the fit. Grey: the official window on the raw trace. Same peak to 0.2 %.

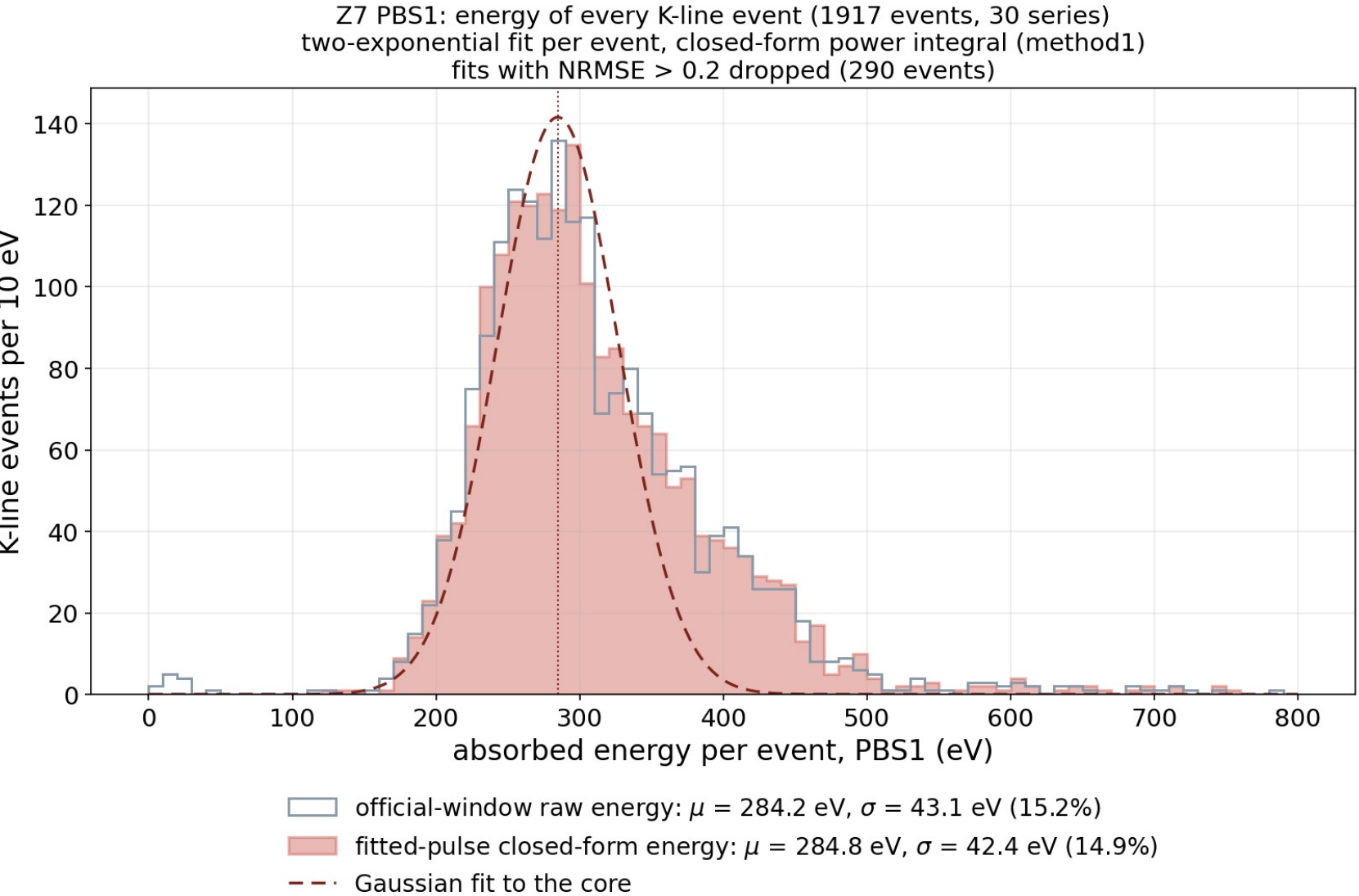
**Peak 285 eV, width 15 % — for a line whose energy is fixed.**

The tail to the right is the events that happened near this channel.

**One channel's width is event position, not noise.**

# 现在看所有事件：一个通道给出又宽又歪的分布

PBS1 ， 1917 个 K 线事件 ， 每个事件一次拟合、一次积分



红色：来自拟合。灰色：官方窗口在原始波形上算的。峰位只差 0.2 %。

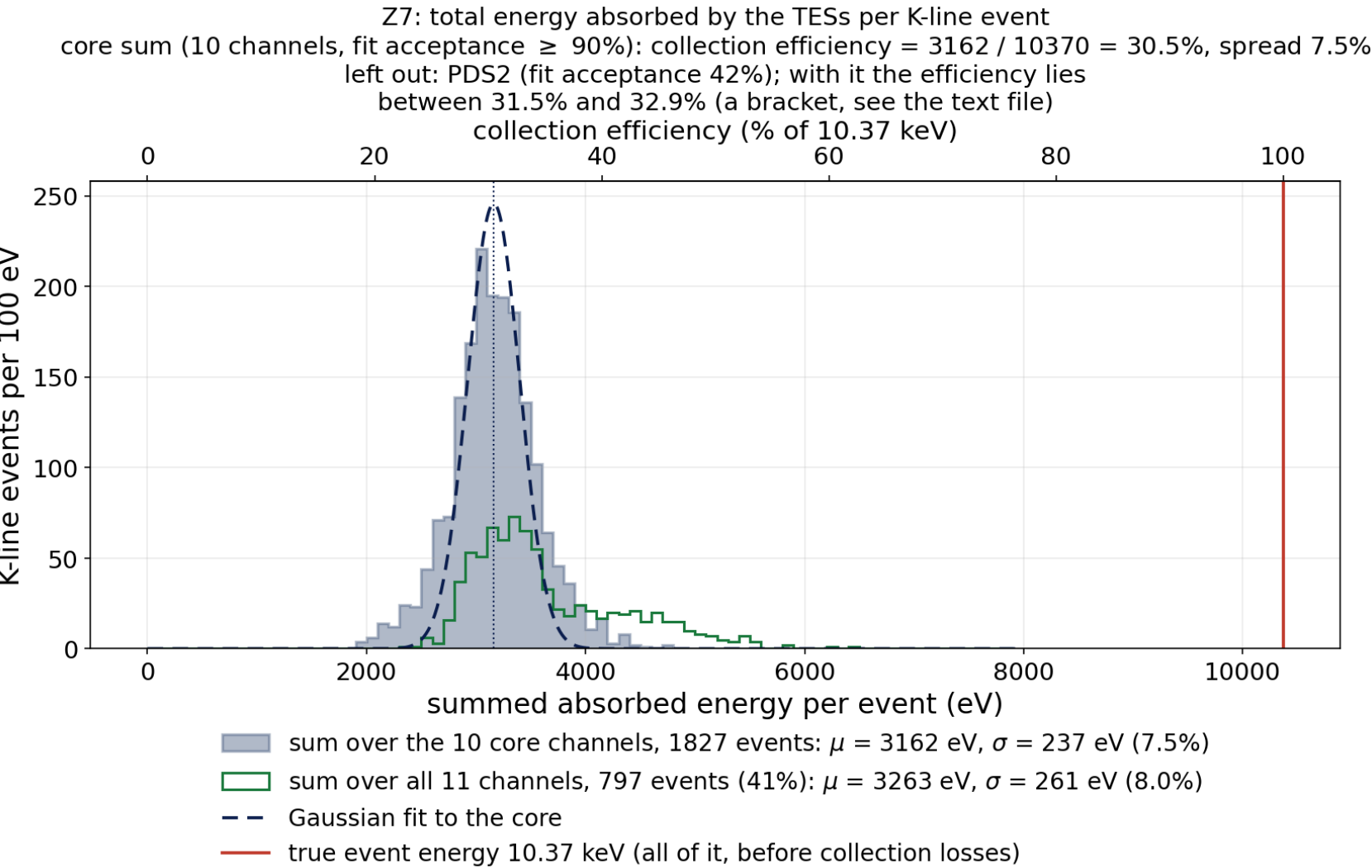
**峰 285 eV ，宽度 15% —— 这可是一条能量固定的谱线。**

往右拖的尾巴，是发生在这个通道附近的事件。

**单通道的宽度是事件位置，不是噪声。**

# Summed over the channels: 32 ± 1 %

the total energy the TESs absorbed per K-line event



Ten channels summed, 1827 events: 3162 eV = 30.5 %, and the width drops to 7.5 % — the position effect cancels.

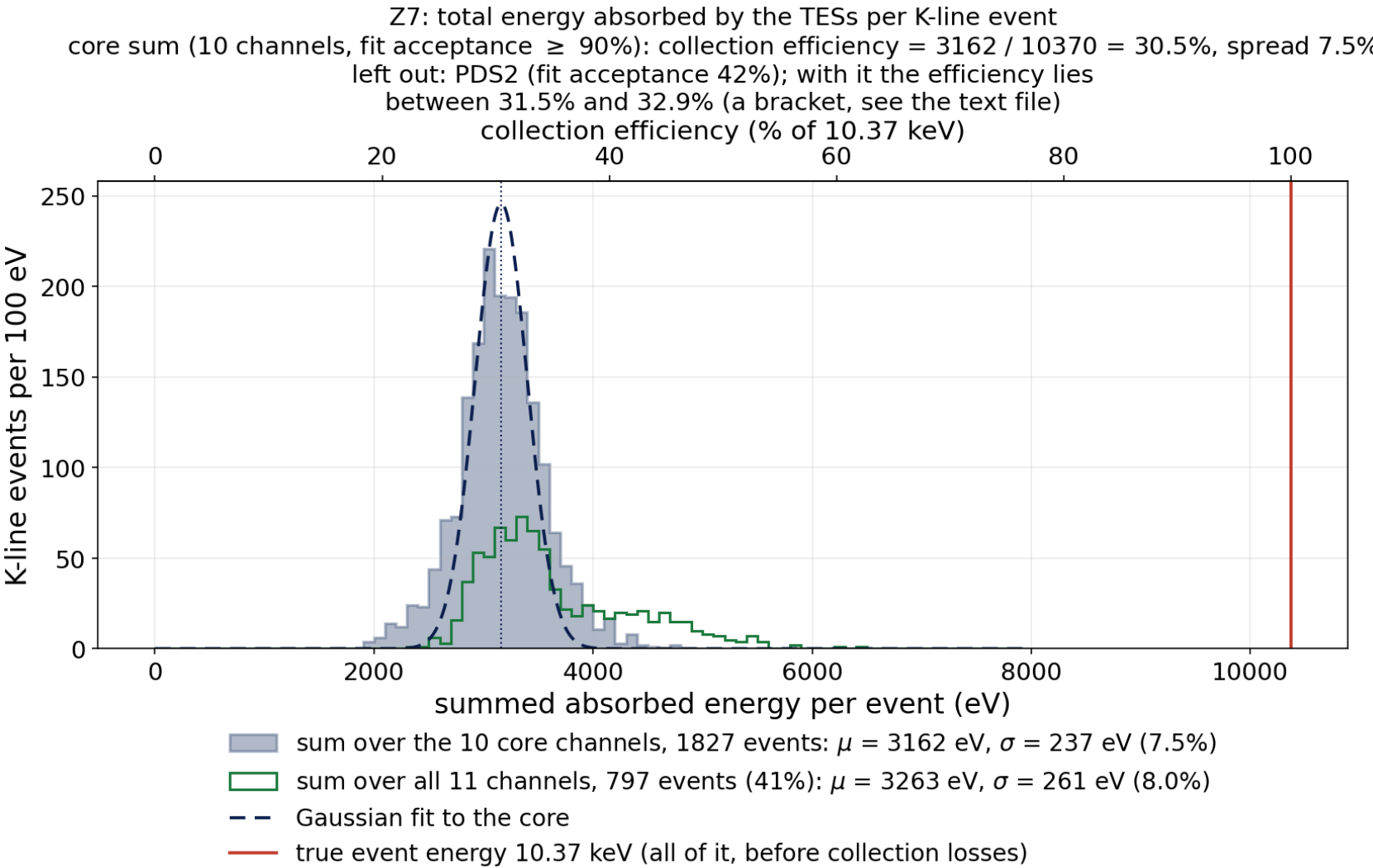
PDS2 fits only 42 % of the events (that slow swing), and those events are not typical; its share can only be bracketed.

Full efficiency 31.5–32.9 %: 32 ± 1 %.

Red line: 10.37 keV — two thirds of the energy never reaches the TESs.

# 各通道求和：32 ± 1%

每个 K 线事件被 TES 吸收的总能量



10 个通道求和，1827 个事件：3162 eV = 30.5%，宽度降到 7.5% —— 位置效应抵消了。

PDS2 只有 42% 的事件能拟合上（那个缓慢起伏），而且这些事件不典型；它的份额只能给一个范围。

完整效率 31.5–32.9%：32 ± 1%。

红线：10.37 keV —— 三分之二的能量根本没到 TES。

# Summary, and what is still open

The current pulse becomes a power pulse through the TES equations; the area under the power is the energy.

The energy is taken from the fitted pulse, so the noise at the peak and the baseline drift do not enter.

**Z7 absorbs  $32 \pm 1\%$  of a 10.37 keV event in its TESs. The CDMS note reports 26–41 % on the R37 CUTE tower.**

Open: Z7 only · the bias of Z7 is not established (0 V assumed) · no dI/dV · PDS2 is the  $\pm 1\%$  · 12.5 % of events fit in no channel.

# 总结，以及还没落实的

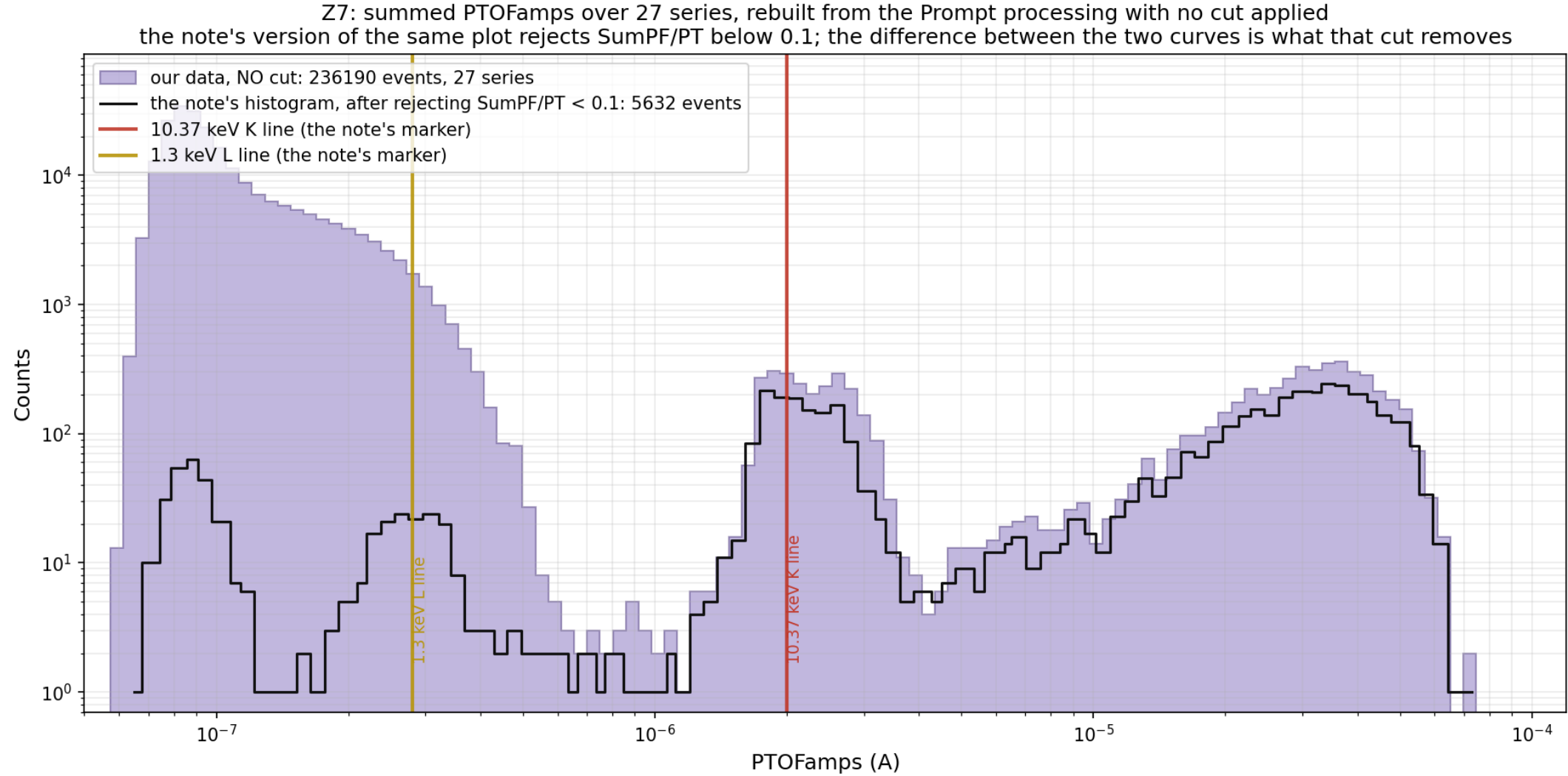
电流脉冲通过 TES 方程变成功率脉冲；功率下面的面积就是能量。

能量用拟合脉冲来算，所以不受峰值噪声和基线漂移的影响。

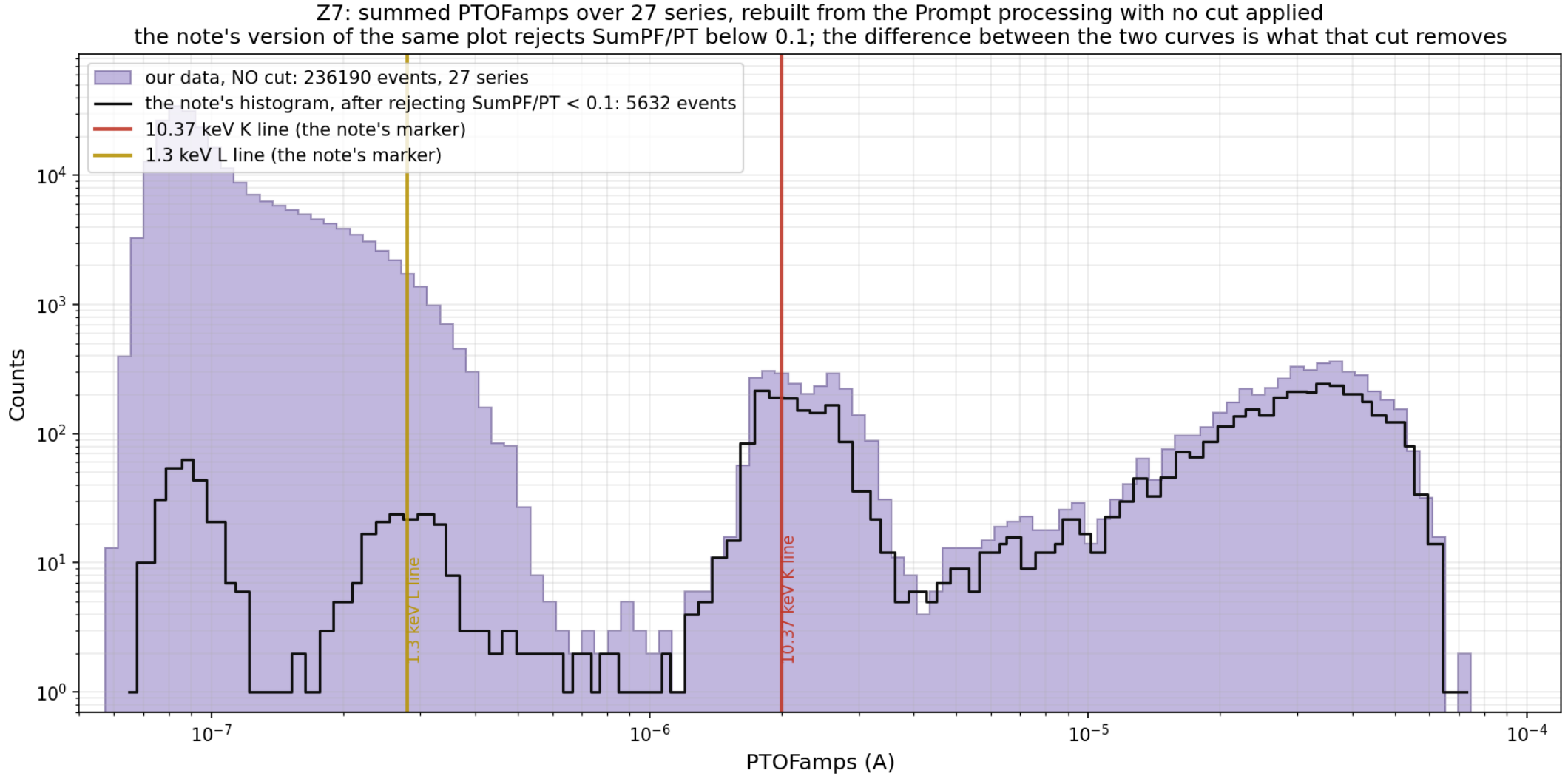
**Z7 把一个 10.37 keV 事件的  $32 \pm 1\%$  吸收进 TES。CDMS 文档在 R37 CUTE tower 上得到的是 26–41%。**

还没落实：只做了 Z7 · Z7 的偏压没有确认（按 0 V 算） · 没有 dI/dV · PDS2 就是那  $\pm 1\%$  · 12.5% 的事件在任何通道都拟合不上。

# Backup — the events



# Backup — 事件谱



# Backup — the two Methods of the CDMS note

$$\delta P = I_0(R_L - R_0) \cdot \delta I + c_2 \cdot R_L \cdot (\delta I)^2 \quad \text{with } c_2 = 2 \text{ (Method 1, used here) and } c_2 = -1 \text{ (Method 2)}$$

The two differ only in the quadratic term: 1.7 % in energy (PBS1, median of 15 events) — far below the 15 % single-channel width and the PDS2 bracket.

The inductor term of the full equations needs L and the loop gain from dI/dV, which do not exist for these series on MSI; it is neglected.

The official Eabs (trigger bin 16383, baseline bins 93..15758, 5-pole 20 kHz prefilter, window −0.5/+1 ms, same formula) is reproduced to 0.2 %; the grey outlines on the histogram slides are it.

# Backup — 文档里的两个 Method

$$\delta P = I_0(R_L - R_0) \cdot \delta I + c_2 \cdot R_L \cdot (\delta I)^2 \quad c_2 = 2 \text{ 是 Method 1 (本报告用的) , } c_2 = -1 \text{ 是 Method 2}$$

两者只有二次项不同：能量差 1.7% （ PBS1 ， 15 个事件的中位数 ）—— 远小于单通道 15% 的宽度和 PDS2 的范围。

完整方程里的电感项需要 L 和 loop gain ，要从 dI/dV 来 ， MSI 上这些 series 没有这个数据；这一项被忽略。

官方 Eabs （ 触发在第 16383 点，基线是 93..15758 点的平均， 5 阶 20 kHz 预滤波，窗口 − 0.5/+1 ms ，同一个公式 ）复现到 0.2% ；直方图页上的灰色轮廓就是它。

# Backup — how the formulas are derived

## 1. From the current to the absorbed power

heat balance of the TES

$$C \dot{T} = P(t) + P_J - P_{bath}$$

over the whole pulse  $\int C \dot{T} dt = 0$ ; the change in bath heat flow is not included

$$E = \int P dt = - \int \delta P_J dt$$

the circuit; at the bias point  $V = I_0(R_L + R_0)$

$$V_{TES} = V - IR_L - L \dot{I}, \quad P_J = IV_{TES}$$

put  $I = I_0 + \delta I$  and subtract the value at the bias point

$$\delta P_J = V \delta I - R_L (2I_0 \delta I + (\delta I)^2) - L I \dot{\delta I}$$

integrate: the  $L$  term gives  $L [I_0 \delta I + (\delta I)^2/2]$ , which is 0 at both ends

$$E = \int \delta P dt, \quad \delta P = I_0(R_L - R_0) \delta I + c_2 R_L (\delta I)^2$$

$c_2 = 1$  from this derivation,  $2$  in Method 1 (used here),  $-1$  in Method 2

## 2. Energy of the fitted pulse

the fit, with  $t$  counted from the pulse start  $t_0$

$$\delta I(t) = A [e^{-t/\tau_f} - e^{-t/\tau_r}]$$

the only integral needed

$$\int_0^\infty e^{-t/\tau} dt = \tau$$

linear term

$$\int \delta I dt = A (\tau_f - \tau_r)$$

quadratic term: expand the square, integrate each exponential

$$\int (\delta I)^2 dt = A^2 \left[ \frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r} \right]$$

so, with  $c_2 = 2$

$$E = I_0(R_L - R_0) A (\tau_f - \tau_r) + 2R_L A^2 \left[ \frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r} \right]$$

PBS1, event 30646:  $A = 1.39 \mu A$ ,  $\tau_r = 136 \mu s$ ,  $\tau_f = 213 \mu s \Rightarrow E = 305.2 \text{ eV}$

## 3. Symbols (Z7 PBS1 values) and assumptions

$I_b$ : bias current    $R_{sh}$ : shunt resistor,  $5.00 \text{ m}\Omega$     $V = I_b R_{sh}$ : bias voltage,  $-1.146 \mu V$     $R_p$ : parasitic resistance,  $14.24 \text{ m}\Omega$

$R_L = R_p + R_{sh}$ : load resistance,  $19.24 \text{ m}\Omega$     $R_0$ : TES resistance at the bias point,  $44.16 \text{ m}\Omega$     $I_0 = V/(R_L + R_0)$ : TES current,  $-18.07 \mu A$

$L$ : inductance (unknown)    $\delta I$ : current change    $A$ : amplitude    $\tau_r, \tau_f$ : rise and fall time constants    $t_0$ : pulse start

Assumptions: detector at 0 V (no NTL effect); the heat flow to the bath is not included;

the  $L$  term cancels only after integration, so  $\delta P(t)$  is not the exact power at each moment; only its integral is used

# Backup — 公式推导和逻辑

## 1. 从电流到吸收的功率

TES 的热平衡

$$C \dot{T} = P(t) + P_J - P_{bath}$$

对整个脉冲积分,  $\int C \cdot dT/dt dt = 0$ ; 不考虑流到热浴的热量的变化

$$E = \int P dt = - \int \delta P_J dt$$

电路方程; 在工作点  $V = I_0(R_L + R_0)$

$$V_{TES} = V - IR_L - L \dot{I}, \quad P_J = IV_{TES}$$

代入  $I = I_0 + \delta I$ , 减去工作点的值

$$\delta P_J = V \delta I - R_L (2I_0 \delta I + (\delta I)^2) - L I \dot{\delta I}$$

积分: 电感项积出来是  $L \cdot [I_0 \cdot \delta I + (\delta I)^2/2]$ , 脉冲两端都是 0

$$E = \int \delta P dt, \quad \delta P = I_0(R_L - R_0) \delta I + c_2 R_L (\delta I)^2$$

$c_2 = 1$  是这个推导的结果,  $2$  是 Method 1 (本报告用),  $-1$  是 Method 2

## 2. 拟合脉冲的能量

拟合函数,  $t$  从脉冲开始的时刻  $t_0$  算起

$$\delta I(t) = A [e^{-t/\tau_f} - e^{-t/\tau_r}]$$

只需要这一个积分

$$\int_0^\infty e^{-t/\tau} dt = \tau$$

线性项

$$\int \delta I dt = A (\tau_f - \tau_r)$$

二次项: 把平方展开, 每个指数分别积分

$$\int (\delta I)^2 dt = A^2 \left[ \frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r} \right]$$

所以, 取  $c_2 = 2$

$$E = I_0(R_L - R_0) A (\tau_f - \tau_r) + 2R_L A^2 \left[ \frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r} \right]$$

PBS1, 事件 30646:  $A = 1.39 \mu A$ ,  $\tau_r = 136 \mu s$ ,  $\tau_f = 213 \mu s \Rightarrow E = 305.2 \text{ eV}$

## 3. 符号 (Z7 PBS1 的数值) 和假设

$I_b$ : 偏置电流    $R_{sh}$ : 分流电阻,  $5.00 \text{ m}\Omega$     $V = I_b \cdot R_{sh}$ : 偏置电压,  $-1.146 \mu V$     $R_p$ : 寄生电阻,  $14.24 \text{ m}\Omega$

$R_L = R_p + R_{sh}$ : 负载电阻,  $19.24 \text{ m}\Omega$     $R_0$ : 工作点的 TES 电阻,  $44.16 \text{ m}\Omega$     $I_0 = V / (R_L + R_0)$ : 工作点的 TES 电流,  $-18.07 \mu A$

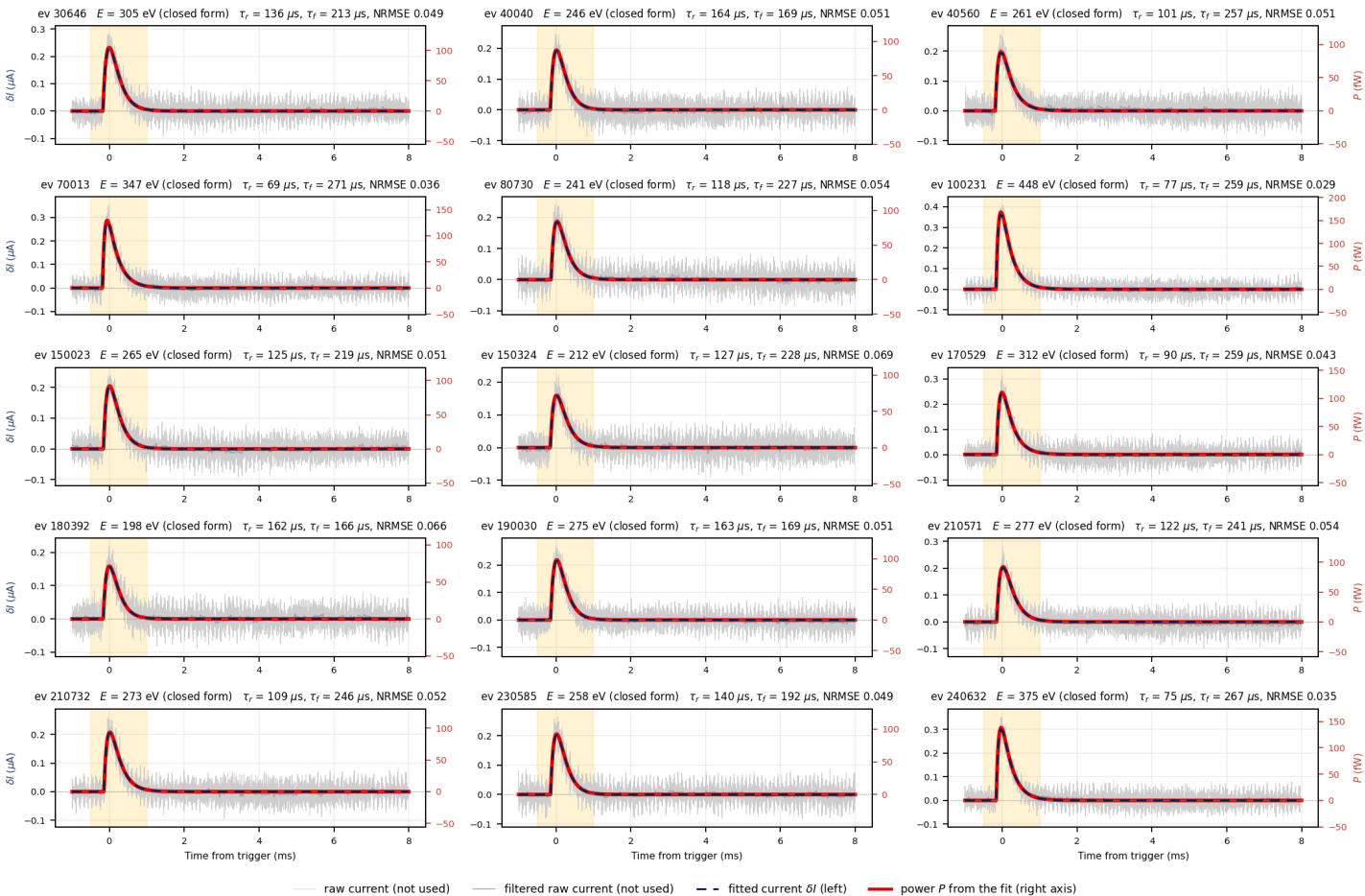
$L$ : 电感 (未知)    $\delta I$ : 电流变化    $A$ : 幅度    $\tau_r, \tau_f$ : 上升和下降时间常数    $t_0$ : 脉冲开始的时刻

假设: 探测器在 0 V (没有 NTL 效应); 不包括流到热浴的热量;

电感项只有积分以后才为 0, 所以  $\delta P(t)$  不是每一时刻的精确功率, 我们只用它的积分

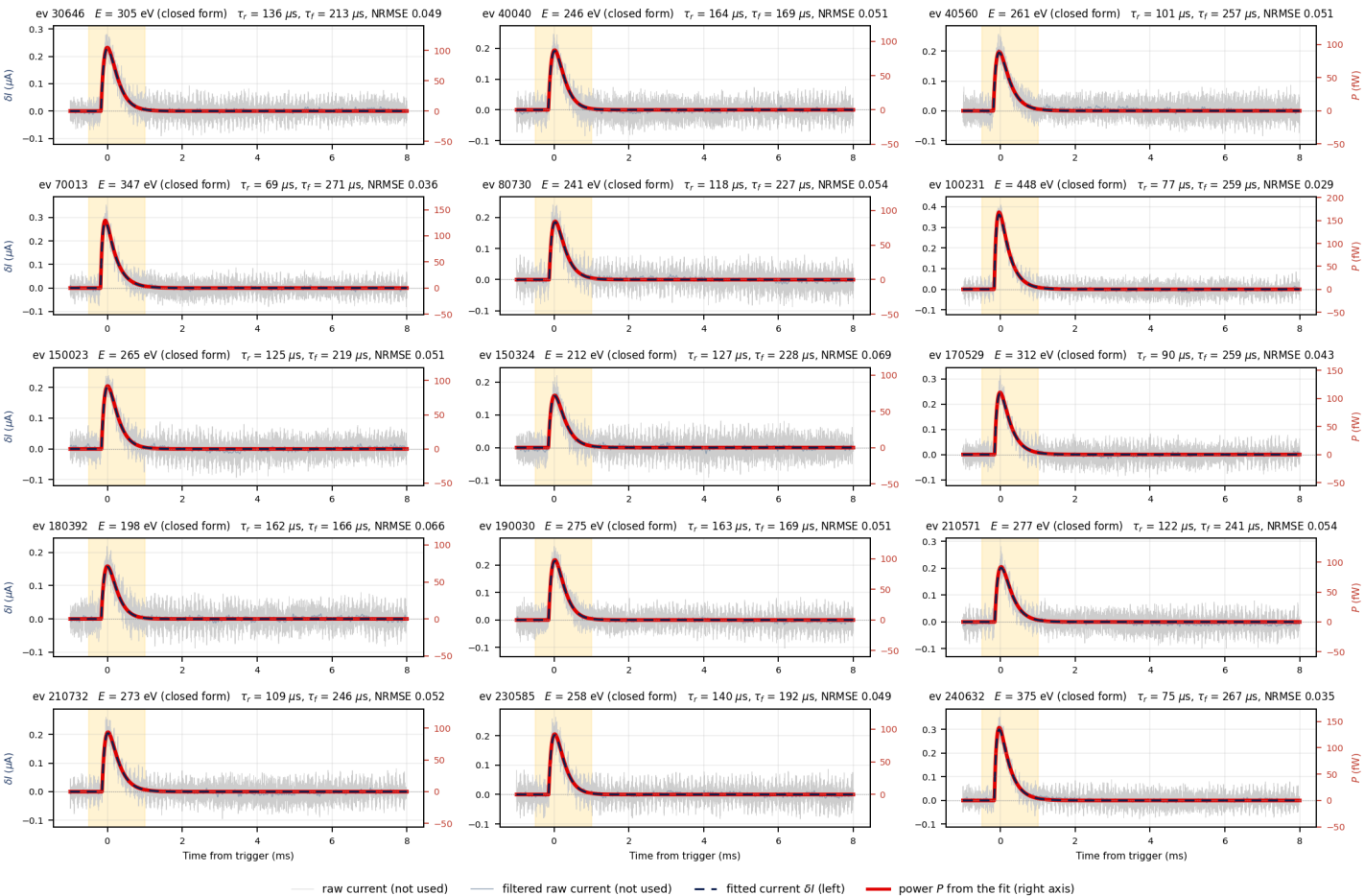
# Backup — 15 events, fitted current and power (PBS1)

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)  
Z7 24260617\_063934, K-line events | method1:  $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * I * (d(t))^2$  | bias point per channel from detectorConfig, ADC/A = 3.145728e+09  
the current is the two-exponential FIT, refitted directly in amperes (model  $A[\exp(-t/\tau_1) - \exp(-t/\tau_2)]$ ), 5 free params, pretrigger free in 16383+3000; the raw trace is drawn only for reference  
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



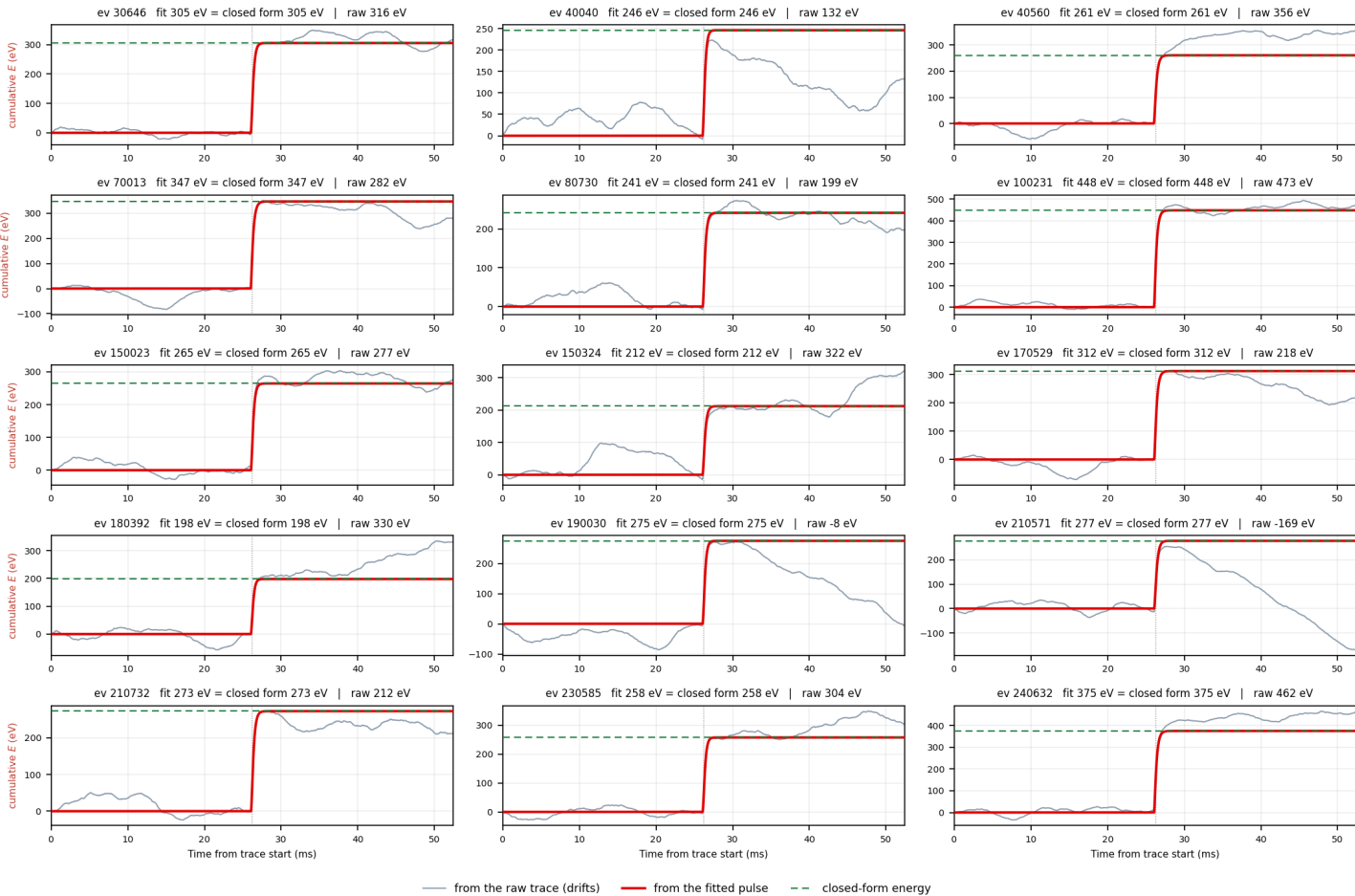
# Backup — 15 个事件，拟合电流和功率（ PBS1 ）

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)  
Z7 24260617\_063934, K-line events | method1:  $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * I * (d(t))^2$  | bias point per channel from detectorConfig, ADC/A = 3.145728e+09  
the current is the two-exponential FIT, refitted directly in amperes (model  $A[\exp(-t/\tau_1) - \exp(-t/\tau_2)]$ ), 5 free params, pretrigger free in 16383+3000; the raw trace is drawn only for reference  
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



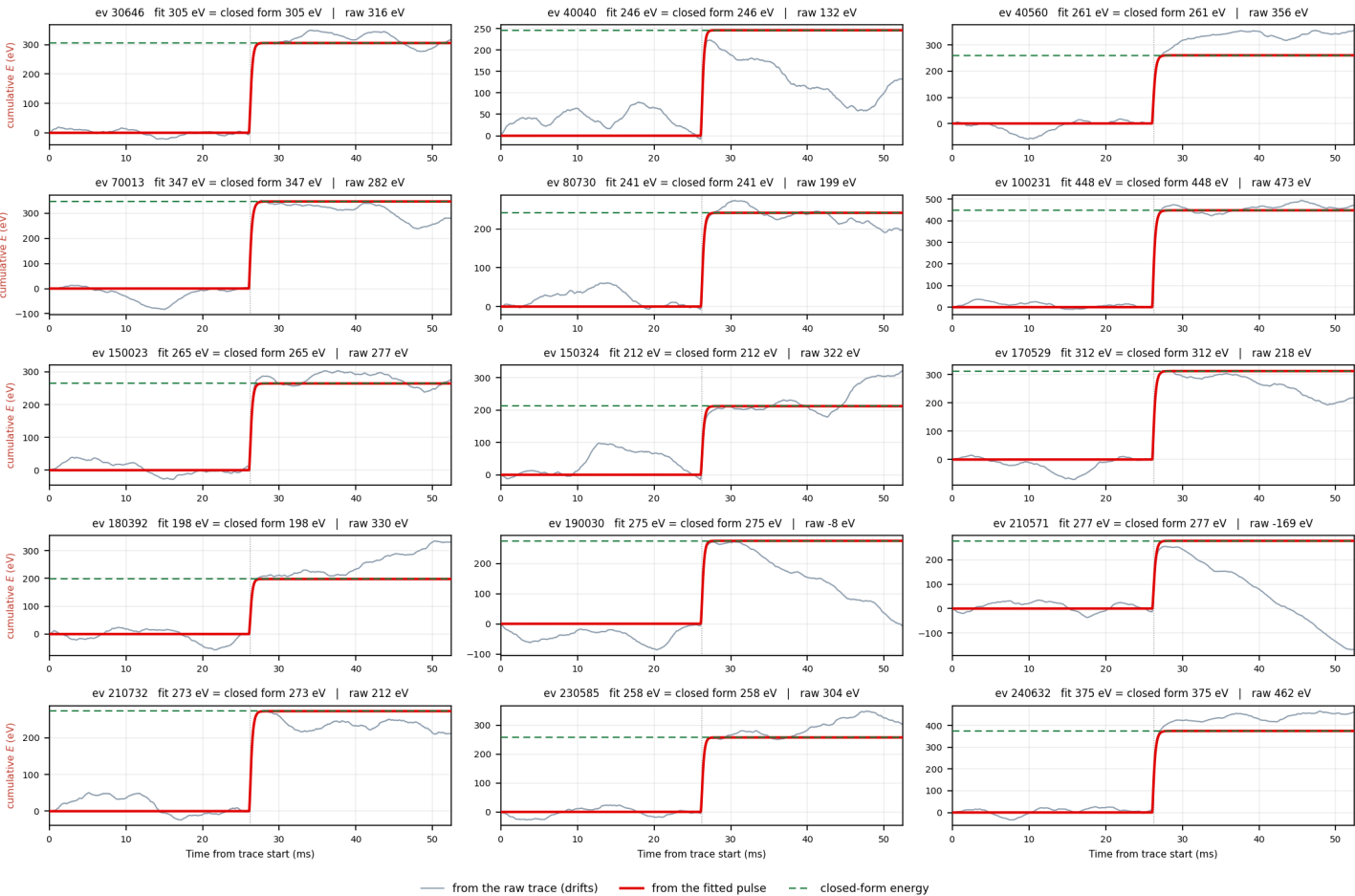
# Backup — 15 events, cumulative energy (PBS1)

Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace  
Z7 24260617\_063934, K-line events | method1:  $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * (d(t))^2$  | bias point per channel from detectorConfig, ADC/A = 3.145728e+09  
the current is the two-exponential FIT, refitted directly in amperes (model  $A[\exp(-t/\tau_1) - \exp(-t/\tau_2)]$ ), 5 free params, pretrigger free in 16383+3000); the raw trace is drawn only for reference  
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



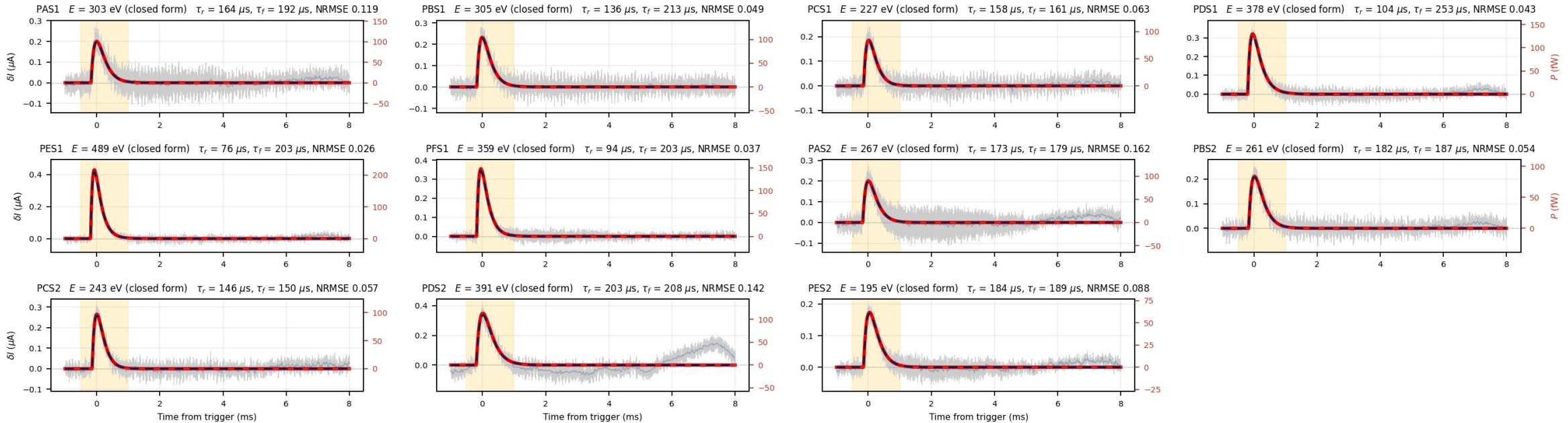
# Backup — 15 个事件，累计能量（PBS1）

Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace  
Z7 24260617\_063934, K-line events | method1:  $P(t) = I_0(R_L - R_0) * d(t) + 2 * R_L * (d(t))^2$  | bias point per channel from detectorConfig, ADC/A = 3.145728e+09  
the current is the two-exponential FIT, refitted directly in amperes (model  $A[\exp(-t/\tau_1) - \exp(-t/\tau_2)]$ ), 5 free params, pretrigger free in 16383+3000); the raw trace is drawn only for reference  
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



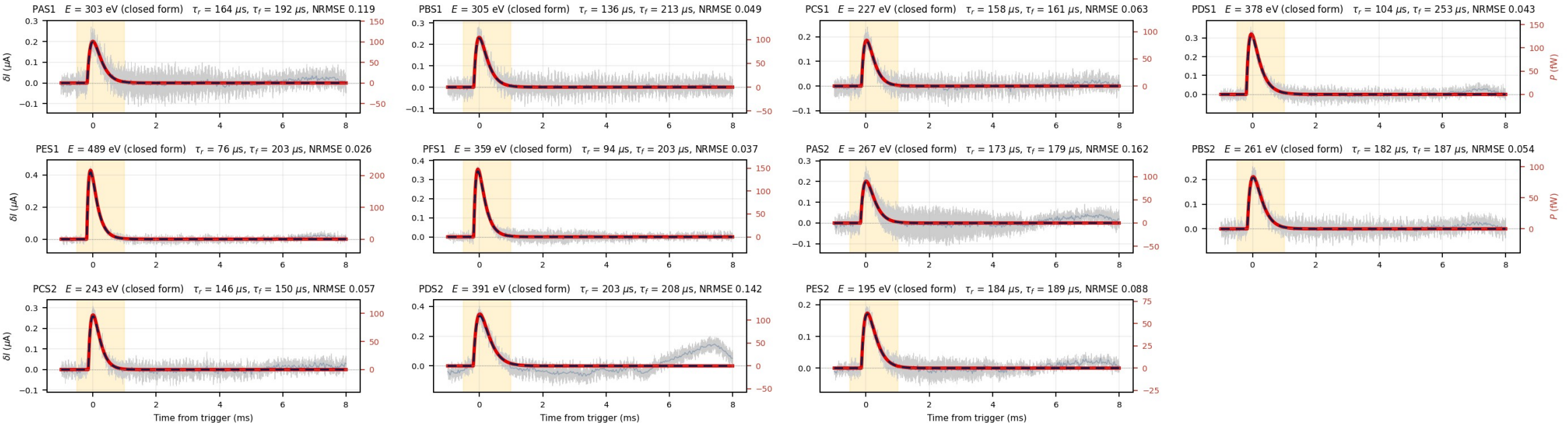
# Backup — event 30646, all channels

Z7 event 30646, all channels: fitted current and power  
sum over channels from the fits: 3419 eV = 33.0% of 10.37 keV (official-window raw sum 3397 eV = 32.8%)  
Z7 24260617\_063934, K-line events | method1:  $P(t) = I_0(R_L - R_0) * dI(t) + 2 * R_L * dI(t)^2$  | bias point per channel from detectorConfig, ADC/A = 3.145728e+09  
the current is the two-exponential FIT, refitted directly in amperes (model  $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$ , 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference  
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



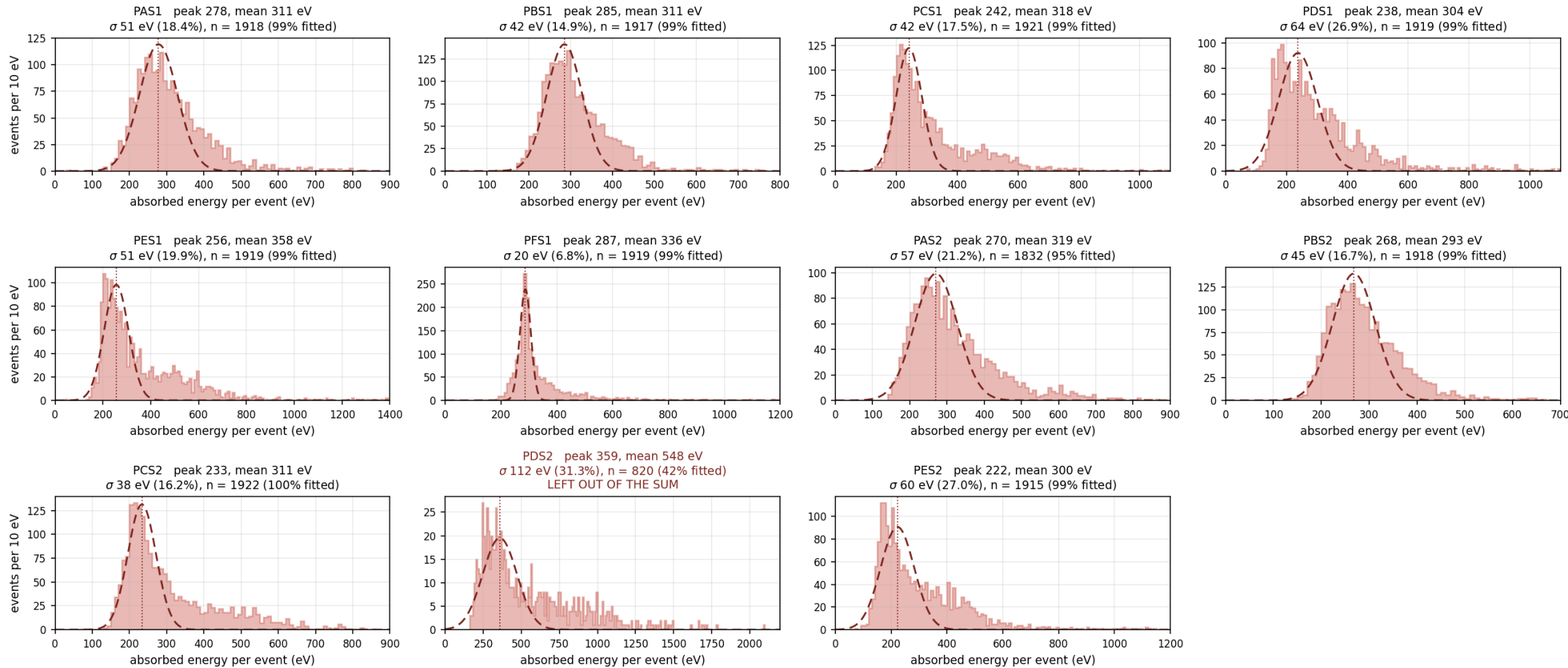
# Backup — 事件 30646 , 全部通道

Z7 event 30646, all channels: fitted current and power  
sum over channels from the fits: 3419 eV = 33.0% of 10.37 keV (official-window raw sum 3397 eV = 32.8%)  
Z7 24260617\_063934, K-line events | method1:  $P(t) = I_0(R_L - R_0) * dI(t) + 2 * R_L * dI(t)^2$  | bias point per channel from detectorConfig, ADC/A = 3.145728e+09  
the current is the two-exponential FIT, refitted directly in amperes (model  $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$ , 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference  
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



# Backup — energy distribution of every channel

Z7: energy of every K-line event, one panel per channel (30 series)  
each entry is one event: its trace fitted with the two-exponential and the power integrated in closed form, method1 quadratic  
the dashed curve is a Gaussian fit to the core, so its centre is the PEAK; these distributions are right skewed, so peaks must not be added (the plain means do add, to 3710 eV = 35.8% of the 10.37 keV K line)



# Backup — 每个通道的能量分布

Z7: energy of every K-line event, one panel per channel (30 series)  
each entry is one event: its trace fitted with the two-exponential and the power integrated in closed form, method1 quadratic  
the dashed curve is a Gaussian fit to the core, so its centre is the PEAK; these distributions are right skewed, so peaks must not be added (the plain means do add, to 3710 eV = 35.8% of the 10.37 keV K line)

